



Pharm*Achieve*

FUNGAL INFECTIONS

Qualifying Exam Preparatory Course

COMPETENCIES COVERED IN THIS PRESENTATION...

**Providing Care:
Clinical Care**

**Providing Care:
Distribution**

**Knowledge
& Expertise**

**Communication
& Collaboration**

**Leadership &
Stewardship**

Professionalism

LEGEND

AIDS Acquired Immunodeficiency Syndrome

C_{max} Peak Concentration

CSF Cerebrospinal Fluid

DI Drug Interaction

GERD Gastroesophageal Reflux Disease

GI Gastrointestinal

HIV Human Immunodeficiency Virus

IA Invasive Aspergillosis

IV Intravenous

LFT Liver Function Test

MOA Mechanism of Action

N/V Nausea/Vomiting

OATP-1B1 Organic Anion Transporting Polypeptide 1B1

OPC Oropharyngeal Candidiasis

PO By Mouth

PPI Proton Pump Inhibitor

SBECD Sulfobutyl Ether Beta-Cyclodextrin

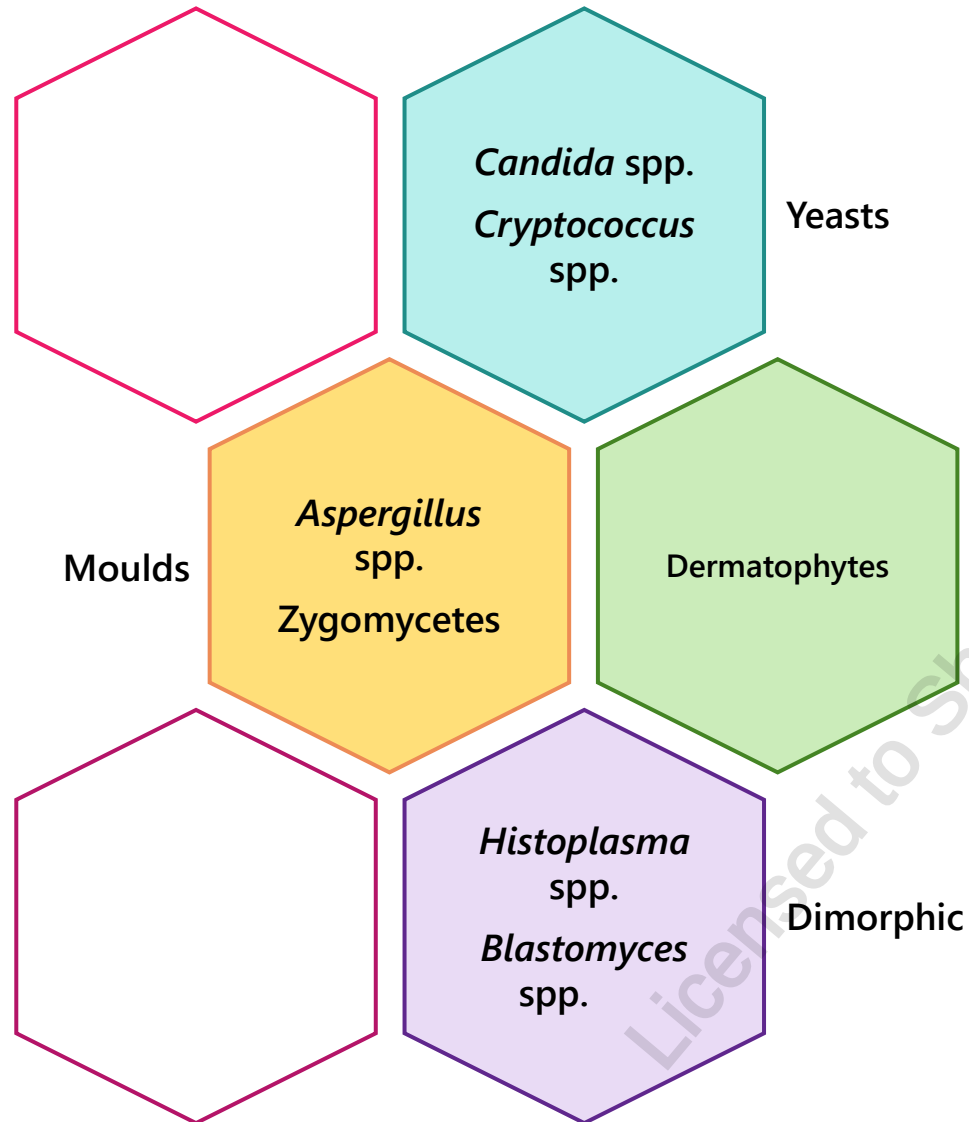
V_d Volume of Distribution

OBJECTIVES

- 1 Identify dermatophyte/tinea infections and understand how to treat them
- 2 Identify signs and symptoms of oral versus esophageal candidiasis, risk factors for oral or esophageal candidiasis
- 3 Understand situations when systemic antifungal therapy may be prescribed instead of topical therapy. What to recommend if fluconazole resistance or failure occurs
- 4 Compare side effects, interactions, and efficacy profiles between different antifungal classes and drugs
- 5 Provide counselling to patients on the various agents

FUNGAL INFECTIONS

FYI



FUNGAL INFECTIONS

Local infection	Invasive Infection
• Vulvovaginal candidiasis • Oral candidiasis • Tinea infections • Tinea Barbae • Tinea Capitis • Tinea Corporis • Tinea Cruris • Tinea Pedis • Tinea Manuum • Onychomycosis	• Esophageal candidiasis • Invasive aspergillosis • Invasive pulmonary aspergillosis • Aspergilloma • Candidemia • Hepatosplenic candidiasis • Pulmonary histoplasmosis/blastomycosis • Cryptococcal meningitis

DERMATOPHYTE/TINEA INFECTIONS: CLASSIFICATION

TINEA BARBAE

- A coarse beard hair, occasionally mustache area
- Typically spread by animals to farm workers
- Unilateral lesions: scaly patches, follicular pustules

TINEA CAPITIS

- Scalp hair follicles and adjacent skin
- Children are commonly affected but also common in low socioeconomic and crowded environments
- Spread through direct contact, animals, contaminated clothing
- Annular patch of scaling and itchy skin along with hair loss

TINEA CORPORIS

- Trunk or limbs excluding face, hands, feet, groin
- Flat, circular scaly spot with central clear portion with raised red border
- Common in athletes with skin-to-skin contact (e.g. wrestlers)

DERMATOPHYTE/TINEA INFECTIONS: CLASSIFICATION

TINEA CRURIS

- Jock itch involves groin area
- Scrotum and penis are spared
- Occurs in men during summer months
- Often a reservoir for infection found on feet
- Bilateral, scaly, red-brown centers with clearly defined raised border

TINEA PEDIS

- Athlete's foot
- White scaling between toes, skin maceration, odour present
- Late-teens, adults
- Occurs from moist environments (e.g. locker rooms, occluded footwear and excessive sweating)

TINEA MANUUM

- Not common
- Palmar/dorsal surface of hands

DERMATOPHYTE/TINEA INFECTIONS: TREATMENT

Non-pharmacologic treatment

- Keep skin clean and dry to discourage fungal proliferation
- Wear loose-fitting clothing for ventilation
- Nonmedicated powders may be used to absorb excess perspiration however avoid cornstarch which promotes fungal proliferation and use caution with talc around neonates/infants or those with respiratory conditions

Pharmacologic Treatment: Topical Agents

Topical Antifungals (clotrimazole 1%, ketoconazole 2%, miconazole 2%, terbinafine 1%, ciclopirox 1%, tolnaftate 1%) for Tinea cruris, corporis, pedis

- Azole duration: 2-4 weeks for cruris; 4 weeks for corporis/pedis
- Topical terbinafine duration: generally, 1-2 weeks (sometimes longer, but generally duration is shorter than azoles)
- Treatment should continue for 1-2 weeks after symptom resolution to lower the risk of recurrence
- Nystatin is NOT effective for tinea infections
- Tinea capitis often requires systemic therapy since topical therapy doesn't penetrate hair follicles. Oral terbinafine is first-line.

DEFINITIONS

Candidiasis

- A fungal infection or mycosis due to *Candida* species
- The most common cause is *Candida albicans*

Oral/esophageal candidiasis

- Fungal infection of the oral mucosa, with *Candida* being primary causative fungi
- The oral infection may extend to the esophagus, causing esophageal candidiasis

EPIDEMIOLOGY

FYI

- *Candida* is found in oral cavity of 30-60% of healthy adults in low numbers and does not cause an infection
- Oral candidiasis is not life-threatening, but can predispose the patient to more serious diseases, like esophageal candidiasis

PATHOPHYSIOLOGY

FYI

Colonization in the oral cavity

Since *Candida* may exist in the oral cavity of healthy adults, individuals can have asymptomatic colonization with no infection, with *C. albicans* being the main colonizing species

Symptomatic infection

- Host and exogenous factors can lead to conversion of asymptomatic colonization to symptomatic infection and disease, causing an oropharyngeal candidiasis (OPC) or esophageal candidiasis
- Normally, the host T-lymphocyte-mediated immune system and neutrophils keep the fungus in check, but with immune conditions like HIV infection or immunocompromised cancer patients, the infection becomes common and apparent

Microbiology

Candida albicans: 70-80%
Non-albicans: 17% which increases to ~50% in immunocompromised patients including following species: *C. glabrata*, *C. tropicalis*, *C. krusei*, *C. guilliermondii*, *C. parapsilosis*

CLINICAL PRESENTATION: ORAL CANDIDIASIS

General descriptions	• Pseudomembranous with “cottage cheese” appearance: thrush • Soft white plaques overlying areas of erythema (red), which can be removed with vigorous rubbing but leave red or bleeding sites • Lesions on tongue
Symptoms	Cotton mouth, loss of taste, may have pain on eating and swallowing, burning sensation on tongue, metallic taste, dysphagia
Signs:	Erythema, white patches/plaques on buccal mucosa, throat, tongue, gums

CLINICAL PRESENTATION: ESOPHAGEAL CANDIDIASIS

General descriptions:	Extension of oral, but esophagus can be the only site involved, distal 2/3 of esophagus, rather than proximal
Symptoms:	• Dysphagia (difficulty of swallowing) • Odynophagia (pain on swallowing) (Hallmark) • Retrosternal chest pain • Epigastric pain (rare)
Signs:	• Fever • Plaques (which could be ulcerated or edematous) • Advanced cases present with narrowing of lumen

LOCAL RISK FACTORS

Steroids and antibiotics	• Suppression of cellular immunity • Alteration of normal flora by broad spectrum antibiotics, which clears the way for Candida and grows with no competition from other microbes
Dentures	• Increased adherence of Candida to acrylic material • Reduced saliva flow • Poor oral hygiene
Anticholinergic side effects	• Dry mouth: decreased saliva production and secretion
Other local factors	• Chemotherapy • Radiotherapy of head/neck • Immunocompromised patients • Organ transplants • HIV/AIDS

SYSTEMIC RISK FACTORS

FYI

Drugs:	• Cytotoxic agents for chemotherapy • Corticosteroids • Immunosuppressants • Drugs that decrease gastric acid by proton-pump inhibition (e.g. omeprazole) • Environmental chemicals (e.g. benzene, pesticides) • Drugs that induce neutropenia
Age:	• Neonates (immature immune system) • Advanced age: Decreased saliva, dentures, less robust immune system, co-morbidities
Diseases:	• HIV/AIDS: Depletion of CD4 T-cells • Diabetes: Higher <i>C. albicans</i> growth in saliva related to increased glucose levels • Malignancies: Leukemia or head/neck cancer • Intensive radiotherapy, chemotherapy • Nutritional deficiencies: Iron, folate, Vit. B1, B2, B6, B12, C

DIAGNOSIS AND DIFFERENTIAL DIAGNOSIS

Diagnosis

- Usually diagnosed by history and physical examination
- In some cases, scraping, a biopsy and endoscopy may be done

Differential Diagnosis

- Thrush: often distinct signs and symptoms make thrush apparent
- Oropharyngeal candidiasis
- Esophageal candidiasis
- GI candidiasis
- GERD (e.g. retrosternal pain)
- Viral esophagitis (herpes simplex, cytomegalovirus)
- Radiation esophagitis

MEDICAL/DRUG RELATED CAUSES OF SIGNS AND SYMPTOMS

Steroids: including inhaled corticosteroids

Xerostomia-inducing agents: including anticholinergic, radiation

Broad spectrum antibiotics: alteration of normal flora (especially in combination with steroids), decreased environmental & nutritional competition for *Candida* causes candidal overgrowth

Chemotherapy: altered rate of mucosal regeneration, xerostomia, neutropenia

PPIs: inhibition of gastric acid allows overgrowth, may inhibit cytotoxic lymphocytes, decreased salivation

GOALS OF THERAPY

- ✓ Cure fungal infections: eliminate signs and symptoms
- ✓ Prevent complications related to fungal infection
- ✓ Minimize adverse effects and manage drug-drug interactions

DRUG THERAPY: TOPICAL AGENTS

Topical therapies

- First line in uncomplicated oral candidiasis with no underlying conditions
- Include oral rinses (suspensions) and troches
- Need to be administered frequently with short contact time
- Need enough saliva to dissolve troches (which may be a problem in patients that have dry mouths)
- May be irritating

Suspensions

- May be better for patients with dry mouth, but the contact time is short

DRUG THERAPY: SYSTEMIC AGENTS

Systemic agents

- Determine if the patient needs systemic or topical therapy based on patient specific factors
- Systemic therapy is needed for esophageal candidiasis
- Those that have trouble with swallowing should be treated with parenteral (IV) formulations
- Imidazole and Triazole antifungal agents fluconazole (IV, PO), itraconazole (PO), voriconazole (IV, PO), posaconazole (PO):
 - Drugs of choice for mucocutaneous candidiasis (e.g. fungal infection on mucous membranes)
- Initial episodes of oropharyngeal candidiasis may be treated with topical therapy, but oral therapy, which is as effective, may be more convenient and better tolerated

SCENARIOS WHERE SYSTEMIC ANTIFUNGAL TREATMENT IS PREFERRED OVER TOPICAL AGENTS

- Patients at high risk for disseminated systemic or invasive candidiasis (e.g. neutropenia due to leukemia and bone marrow transplantation)
- Patients who cannot tolerate topical agents due to dry mouth or swallowing issues
- Patients with oral/esophageal candidiasis refractory to topical treatment
- Severe odynophagia

DRUG THERAPY: OROPHARYNGEAL CANDIDIASIS

Primary therapy	Clotrimazole	Topical azoles compounded to troches
	Nystatin	Oral polyenes suspension or pastilles - preferred treatment options
	Fluconazole	Oral azoles tablet (for more severe cases)
Alternative therapy	Itraconazole	Oral imidazole solution
	Voriconazole/Posaconazole: refractory cases	Oral azole suspension
	Amphotericin B: severe or refractory cases	IV
	Echinocandins: severe or refractory cases	IV

DRUG THERAPY: ESOPHAGEAL CANDIDIASIS

Primary therapy	Fluconazole*	Oral tablet (preferred)
	Itraconazole	Oral solution
Alternative therapy (severe or refractory cases)	Voriconazole	Oral/IV
	Anidulafungin	IV
	Caspofungin	IV
	Micafungin	IV
	Amphotericin B	IV
	Posaconazole	Oral

* IV alternative available if oral route can't be used (e.g. inability to swallow)

SUMMARY OF ANTIFUNGAL AGENTS

Azoles	Fluconazole, itraconazole, voriconazole, posaconazole, isavuconazole, ketoconazole MOA: inhibit CYP450 needed in the biosynthetic pathway of ergosterol, an essential molecule in fungal cell membranes
Echinocandins	IV only: caspofungin, micafungin, anidulafungin MOA: inhibit synthesis of an essential cell wall component of <i>Candida</i> , non- <i>Candida</i> & <i>Aspergillus</i> (Beta 1,3-glucan)
Polyenes	Nystatin, amphotericin B MOA: Interact with fungal membrane to produce channels through membrane, causing leakage of essential cell contents

AZOLES

- There are 2 classes of azole antifungal agents:
 - Imidazoles: 2-N azole ring (e.g. clotrimazole, ketoconazole)
 - Triazoles: 3-N azole ring, better pharmacokinetic properties, including fluconazole, itraconazole, voriconazole, posaconazole, isavuconazole
- Fluconazole and itraconazole are older
- Voriconazole, isavuconazole and posaconazole are newer and made to:
 - Overcome the limited efficacy of fluconazole against *Aspergillus* and other moulds along with non-albicans *Candida* spp *C. krusei* and *C. glabrata*.
 - To improve absorption, tolerability, and safety compared to Itraconazole

MOA: Inhibit CYP450 needed in the biosynthetic pathway of ergosterol, which is an essential molecule in fungal cell membranes

AZOLES

- Drug interactions (DI):
 - The most important DI is ↑ of the following drugs due to the inhibition of CYP450s: (3A4: itraconazole, posaconazole, voriconazole), (3A4, 2C9/19: fluconazole, voriconazole)
 - Cyclosporine, tacrolimus, sirolimus, most calcium channel blockers, most benzodiazepines, most statins, steroids, warfarin, rifampin
 - Itraconazole and posaconazole are inhibitors of P-glycoprotein (efflux pump which inhibits absorption of several drugs)
 - Inducers of CYP450s (carbamazepine, phenobarbital, phenytoin, rifampin, and rifabutin) ↓ azole concentrations
- Side effects: N/V, LFTs (liver function tests) increase (5-10%), gynecomastia (particular ketoconazole at higher doses for long duration, except fluconazole where gynecomastia is not an issue)

AZOLE DRUG TABLES

Fluconazole	• Water soluble and very well absorbed; not affected by food or GI pH • 12% protein bound; 80% excreted unchanged in urine → Dose adjust for renal dysfunction • Hepatic CYP2C9 plays role in its metabolism • Penetrates into body fluids and CSF • Side effects: GI upset, hepatitis, increased LFTs • Interactions: least drug-interacting agent compared to all azoles but renally excreted • Its efficacy is decreased by rifampin • Increases toxicity of cyclosporin, contraceptives, prednisone, sulfonylureas, theophylline, and warfarin
Itraconazole	• Water insoluble, bioavailability variable after oral administration • Capsule best absorbed with food, oral solution better absorbed on empty stomach (fasting state) with higher bioavailability than capsule • Solubilized in oral solution, or IV form suspended by cyclodextrin molecule (inactive oligosaccharide molecule acts like a vehicle for the hydrophobic drug) • Equal amounts of native drug, and hydroxyitraconazole metabolite in plasma. This metabolite has equal efficacy, both are 99% protein bound • Metabolized by the liver, stronger inhibitor of CYP3A4 • Does not penetrate CSF • Side effects: N/V, hepatotoxicity, congestive heart failure, pulmonary edema

AZOLE DRUG TABLES

Voriconazole

- Poorly soluble, even though structurally similar to fluconazole
- More efficacious versus fluconazole for invasive aspergillosis and non-*albicans* Candida
- Orally administered, either tablet or solution
- Bioavailability is 96%, reduced 22% by food
- Solubility in IV achieved by cyclodextrin (SBECD: sulfobutyl ether beta-cyclodextrin)
- 56% protein bound, good V_d (volume of distribution)
- Metabolized in the liver, mainly by CYP2C19, CYP3A4, and lesser extent by CYP2C9
- Penetrates into vitreous and aqueous humour
- 2% in urine, main metabolite is inactive N-oxide
- Affected by CYP2C19 genetic variability (19% Asians and 2% Caucasians are poor metabolizers)
- Drug interactions: sirolimus, tacrolimus, cyclosporine, midazolam, rifabutin/rifampin, ergots
 - ↓voriconazole levels by carbamazepine, rifampin, barbiturates, phenytoin
- Side effects: visual field transient disturbances (20-23%) presenting as altered perception, acuity, photophobia (subsides in 30-60 min after giving dose); visual and auditory hallucinations, hepatitis

AZOLE DRUG TABLES

Posaconazole	• Has only been studied in a very selected patient group for invasive aspergillosis (IA) prevention and treatment along with treatment/prevention of zygomycetes • Hematopoietic stem-cell transplant recipients with graft-versus-host disease • Only as suspension for oral use • Pharmacokinetics: • Absorption increased by food, especially fatty meals (area under the curve and Cmax increased 2-4 times by a fatty meal), better absorbed in divided daily doses • Not affected by antacids • > 98% protein bound, good tissue penetration • Metabolized in the liver by glucuronidation and excreted mostly in the bile and urine • Moderate inhibitor of CYP3A4
Ketoconazole	• Least effective of all azoles due to poor bioavailability • 62-82% cure rate compared to > 85% with itraconazole or fluconazole • Low pH needed for absorption, therefore acid suppression decreases absorption • In practice, not systemically used because of: greater toxicity, lower efficacy, drug interactions • Replaced by newer azoles such as fluconazole and itraconazole • Interactions: most interacting of all azoles, since it is a potent 3A4 inhibitor • Side effects: gynecomastia, nausea, vomiting, abdominal pain, itching, headache, hepatotoxicity, endocrine effects, lichenoid mucosa reactions

ECHINOCANDINS PROPERTIES

FYI

- Semisynthetic lipopeptides
- In candidemia trials, all echinocandins were as efficacious as fluconazole, amphotericin B, or lipid formulation of amphotericin B
- Resistance is infrequent in *Candida spp.* (common in *C. auris*)
- Low oral bioavailability, therefore available only as parenteral formulations
- Mechanism of action:
 - Inhibit synthesis of an essential cell wall component of *Candida* and *Aspergillus* (Beta 1,3-glucan)
 - Effective against *Candida* and *Aspergillus* because of the high concentration of Beta 1,3-glucan in their cell walls, inactive against other fungi
 - Not effective against zygomycetes, *Cryptococcus neoformans*, *Fusarium* species
- Drug interactions: none are major inducers, substrates, and inhibitors of CYP450 or P-glycoprotein, therefore have low possibility for major drug interactions.

ECHINOCANDINS PROPERTIES

Caspofungin and Micafungin:

- Because of unique mechanism of action, can be used in combination with azoles and polyenes
- Can be used more in immunocompromised patients
- Narrow spectrum of activity
- Fungicidal against most *Candida* species
- Side effects:
 - Very few toxic side effects, because target of the cell wall component is not found in mammalian cells, therefore safe to prescribe in patients with underlying diseases, and at the same time with many drugs
 - ↑ LFTs, headache, histamine release (rash), phlebitis (due to IV administration), fever
- Advantages:
 - As efficacious as fluconazole in *Candida esophagitis*
 - Resistance of *Candida* to echinocandins has been rare so far
 - Extremely safe antifungals, relative to other antifungal classes
 - Do not require dose adjustments in renal impairment (dose adjust still required for hepatic impairment)
- Disadvantages: higher cost, more inconvenient because of IV administration

ECHINOCANDINS PROPERTIES

Pharmacokinetics: **FYI**

- Low oral bioavailability
- High protein binding
- Low urine concentrations
- Low CSF, brain, and ocular penetration
- Degraded in the liver by N-acetylation and hydrolysis

Drug interactions:

- May have to increase the dose of caspofungin when taken at the same time with some enzyme inducers such as phenytoin, efavirenz, nevirapine, dexamethasone and carbamazepine
- Caspofungin uses OATP transporter (organic anion-transporting polypeptide), which also transports bile, rifampin, pravastatin, irinotecan, and cyclosporine
- Rifampin induces the hepatic uptake of caspofungin by OATP-1B1, resulting in decreased steady/state plasma concentrations, therefore have to increase the dose of caspofungin

ECHINOCANDINS

Echinocandin	Drug interactions	Side effects
Micafungin	Increased levels of sirolimus and nifedipine	• N/V • Headache • Flushing (due to histamine release) • Infusion-related adverse effects such as pruritus, erythema, and pain • Elevations in liver enzymes • Anaphylaxis
Caspofungin	• Decreased caspofungin levels by rifampin, efavirenz, phenytoin, dexamethasone, carbamazepine • Increased caspofungin levels by cyclosporine • Increased tacrolimus levels	
Anidulafungin	-	• Same as above except ↑ LFTs is less than all other echinocandins

POLYENES

Nystatin	• The only non-absorbable topical form • Effective in oral candidiasis but less effective compared to oral azoles (fewer cures and more relapse with nystatin) • Interacts with fungal membrane to produce channels through the membrane, causing leakage of essential cell contents
Amphotericin B	• Widest spectrum of activity among all antifungal agents • Lipid-based formulation provides better safety profile (less nephrotoxicity) compared to plain amphotericin, and better pharmacokinetic parameters • Side effects: N/V, diarrhea (higher with higher doses, infusion-related side effects can be prevented by slowing down the infusion rate, treat with diphenhydramine and acetaminophen) • Monitor electrolytes due to occurrence of hypokalemia, hypomagnesemia, hypophosphatemia • Monitor renal function, since it is a nephrotoxic drug; do not combine with other nephrotoxic drugs

MONITORING

Efficacy Endpoints		
Parameter	Degree of Change	Timeframe
Signs and symptoms	Improved and/or Resolved No new lesions appearing	1-4 weeks for clearing of lesions (depending on infection site and specific treatment used) Erythema and pruritus usually improve in a few days

Safety Endpoints		
Parameter	Degree of Change	Timeframe
Adverse effects from drug therapy and complications	Prevent	Duration of therapy

TAKEAWAYS

Tinea infections:

- **Non-pharmacologic treatment** is important (keep skin clean and dry, wear loose-fitting clothes)
- **Pharmacologic treatment:**
 - Topical agents (e.g. **topical azoles, terbinafine, tolnaftate, ciclopirox**) are 1st line therapy for tinea pedis, tinea corporis and tinea cruris
 - Duration of terbinafine 1% is usually shorter than that of topical azoles
 - Systemic therapy is required for tinea capitis since topical therapy does not penetrate hair follicles
 - Nystatin is **NOT** effective in tinea infections because dermatophytes are resistant to nystatin

Candida infections:

- Oral candidiasis: **Nystatin oral suspension** is the first line
- Esophageal candidiasis: systemic agents are used

REFERENCES

1. Goldstein AO, Goldstein BG. Dermatophyte (tinea) infections. UpToDate. www.uptodate.com.
2. Kauffman CA, Vazquez JA. Esophageal candidiasis in adults. UpToDate. www.uptodate.com.
3. Kauffman CA, Vazquez JA. Oropharyngeal candidiasis in adults. UpToDate. www.uptodate.com.
4. Miller PF. Fungal Skin Infections. In: Minor Ailments. Ottawa. ON: Canadian Pharmacists Association. www.myrxtx.ca.
5. Antifungal agents for common outpatient paediatric infections. Canadian Paediatric Society. <https://cps.ca/documents/position/antifungal-agents-common-infections>.
6. Candida infections of the mouth, throat, and esophagus. Centers for Disease Control and Prevention. <https://www.cdc.gov/fungal/diseases/candidiasis/thrush/index.html>.
7. Taylor M, Brizuela M, Raja A. Oral Candidiasis. In: StatPearls. <http://www.ncbi.nlm.nih.gov/books/NBK545282/>.
8. Wlock K. Oral Candidiasis. In: Minor Ailments. Canadian Pharmacists Association. <https://www.myrxtx.ca>.
9. Therapeutic Guidelines For Opportunistic Infections Candidiasis. https://bccfe.ca/wp-content/uploads/2024/04/bc-cfe_therapeutic_guidelines_for_opportunistic_infections-candidiasis-mar-2023.pdf.
10. Mohamed AA, Lu XL, Mounmin FA. Diagnosis and Treatment of Esophageal Candidiasis: Current Updates. *Can J Gastroenterol Hepatol*. 2019;2019:3585136. Published 2019 Oct 20. doi:10.1155/2019/3585136.
11. Pappas PG, Kauffman CA, Andes DR, et al. Clinical Practice Guideline for the Management of Candidiasis: 2016 Update by the Infectious Diseases Society of America. *Clin Infect Dis*. 2016;62(4):e1-e50. doi:10.1093/cid/civ933.
12. Thomas B, Falk J, Allan GM. Topical management of tinea pedis. *Can Fam Physician*. 2021;67(1):30. doi:10.46747/cfp.670130.
13. Leung AK, Barankin B, Lam JM, Leong KF, Hon KL. Tinea pedis: an updated review. *Drugs Context*. 2023;12:2023-5-1. Published 2023 Jun 29. doi:10.7573/dic.2023-5-1.
14. Weinstein A, Berman B. Topical treatment of common superficial tinea infections. *Am Fam Physician*. 2002;65(10):2095-2102.

CHANGE LOG

April 2025

- Slide 8: Added 'palmar' for tinea manuum (more common than dorsal)
- slide 18: Added info for steroids and xerostomia
- Slide 23-24: Changed to itraconazole 'solution'
- Slide 24: Removed itraconazole capsule as they are less effective
- Slide 26: Added isavuconazole under triazoles
- Slide 36: Updated reference slide

July 2025

- Slide 26: Updated azole coverage spectra
- Slide 27: Amended voriconazole CYP enzymes

October 2025

- No content changes made

February 2026

- Slide 2: Updated NAPRA competencies
- Added FYI on a few slides
- Slide 9: Updated information on pharmacologic treatment and added tolnaftate 1% to list of topical agents
- Slide 23: Minor update
- Slide 24: Added note about IV fluconazole as an alternative
- Slide 36: Added monitoring parameters
- Slide 38: Updated references